# Manuel Rodríguez Villegas

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## Profile

AI engineer with a strong mathematical background, building systems that see, learn and act: drones that fly from camera pixels, mapping robots and AI agents that assist engineers on their day-to-day work.

## Education

**Comillas Pontifical University, ETSI ICAI** — Madrid, Spain
*Master's Degree in Artificial Intelligence; GPA: 9.08/10; Class Representative* — *2025 – 2026*

**Cornell University** — Ithaca, NY, USA
*Exchange Student - Electrical & Computer Engineering* — *Jan – May 2025*

**Comillas Pontifical University, ETSI ICAI** — Madrid, Spain
*BE in Mathematical Engineering and AI; GPA: 8.75/10* — *2021 – 2025*

## Experience

**AUDI AG** — Ingolstadt, Germany
*AI Engineering Intern* — *Jul – Nov 2026*
- Building an internal AI coworker on Azure for requirements engineers: it queries a graph database of 500,000+ entries fast, reliably and with minimal tokens, classifies requirements and detects conflicts and duplicates.
- Developing 3D defect classification models under limited data and strong confounding.

**Comillas Pontifical University** — Madrid, Spain
*Teaching Assistant* — *Sep 2025 – Apr 2026*
- Contributed to deployment of a GPU-based AI cluster to support student workloads in deep learning courses.

**Azzulei Technologies** — Madrid, Spain
*Computer Vision Intern* — *Jun – Aug 2025*
- Built a real-time player-and-ball tracking pipeline (YOLOv8, ByteTrack, optical flow) for automated sports camera control, robust to a small, fast ball and occlusions.

**Imperial College London** — Remote (London, UK)
*Research Assistant* — *Jun – Aug 2024*
- Modelled nonlinear dynamical systems (Lotka–Volterra, SIR, Lorenz) with Neural ODEs and PINNs.

**Endesa** — Madrid, Spain
*Energy Data Analyst Intern* — *Jun – Aug 2023*
- Forecasted daily Iberian ancillary service costs using MLP and regression models (MAPE $< 10\%$).

## Selected Research

**Master's Thesis: Vision-Only Autonomous Drones Trained with Differentiable Simulation** — 2026
- Training an FPV drone to avoid obstacles from RGB images alone with SHAC, backpropagating through differentiable JAX physics and a gaussian-splat renderer into a CNN + GRU policy.
- Reached 92.5% success on 1,024 evaluation corridors after just 0.9 hours of training on a single GPU.

**TopoSIGMA: Spectral Information-Geometric Mapping Architecture** — 2025
- Designed an online, training-free ROS 2 mapping system separating spectral place discovery from covariance-aware loop closure, using DINOv2 descriptors and directional node distributions.
- Manuscript in preparation for submission to Robotics and Autonomous Systems.

## Skills

**Software**: Python (advanced), PyTorch, JAX, ROS 2, Git, Docker, K8S, Azure, LangGraph, n8n
**AI & ML**: Agentic AI, Computer Vision, Geometric AI, Gaussian Splatting, Differentiable Simulation, NLP, DRL
**Mathematics**: Linear Algebra, Probabilistic AI, Optimization, Differential Geometry, Statistics
**Languages**: Spanish (Native), English (Fluent), German (Beginner)

## Honors & Awards

**Bending Spoons Hackathon Winner** — Milan, Italy | 2026
*Selected among Spain's top 20 tech students for First Ascent and winner of the event's hackathon.*

**10th Smart Industry Hackathon Winner** — Madrid, Spain | 2025
*Built a virtual assistant with recommendations for train operators, in a hackathon featuring Kearney and CAF.*

**UNIJES Social Tech Challenge Winner** — Bilbao, Spain | 2024
*Voice-controlled wheelchair project*

**Academic Excellence Scholarship** — Madrid, Spain | 2021, 2023, 2024
*Top-scoring students across Community of Madrid*

**Baccalaureate Honorific Mention** — Madrid, Spain | 2021
*2nd highest GPA in Community of Madrid*